

PLC Training Manual

Introduction

A Programmable Logic Controller (PLC) is a rugged industrial computer designed to automate manufacturing processes.

Main Components

- CPU
- Power Supply
- Input Modules
- Output Modules
- Communication Modules

PLC Scan Cycle

1. Input Scan
2. Program Execution
3. Output Scan
4. Housekeeping

Programming Languages

- Ladder Logic
- Function Block Diagram
- Structured Text
- Sequential Function Chart

Applications

• Conveyor Systems • Robotic Automation • Packaging Machines • Water Treatment Plants • Assembly Lines

Advantages

• High Reliability • Fast Processing • Easy Maintenance • Expandable Architecture • Real-Time Control